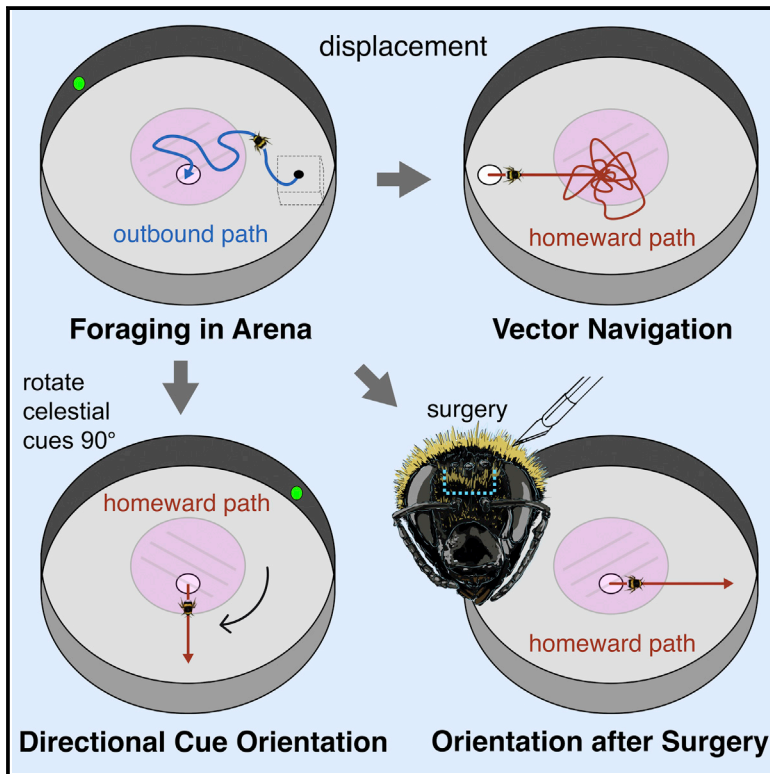


Current Biology

Vector navigation in walking bumblebees

Graphical abstract



Authors

Rickesh N. Patel, Julian Kempenaers, Stanley Heinze

Correspondence

rickesh.patel@biol.lu.se

In brief

Bees perform impressive vector navigation feats while foraging over long distances. Patel et al. show that bumblebees also use path integration over short distances while walking in the laboratory. This vector navigation behavior is highly amenable and resilient to anesthetization and surgeries commonly required for electrophysiological investigations.

Highlights

- Bumblebees use robust vector navigation over short distances while walking
- Walking bees estimate correct vector distances and orient to visual cues
- Bees exhibit search behaviors typical for a path-integration-based strategy
- Walking vector navigation behavior is resilient to anesthetization and surgery



Article

Vector navigation in walking bumblebees

Rickesh N. Patel,^{1,3,5,*} Julian Kempenaers,¹ and Stanley Heinze^{1,2,4}¹Lund Vision Group, Department of Biology, Lund University, Lund, Sweden²Nano Lund, Lund University, Lund, Sweden³Twitter: @RickeshNPatel⁴Twitter: @stanley_heinze⁵Lead contact

*Correspondence: rickesh.patel@biol.lu.se

<https://doi.org/10.1016/j.cub.2022.05.010>

SUMMARY

Path integration is a computational strategy that allows an animal to maintain an internal estimate of its position relative to a point of origin. Many species use path integration to navigate back to specific locations, typically their homes, after lengthy and convoluted excursions. Hymenopteran insects are impressive path integrators, directly returning to their hives after hundreds of meters of outward travel. Recent neurobiological insights have established hypotheses for how path integration vectors could be encoded in the brains of bees, but clear ways to test these hypotheses in the laboratory are currently unavailable. Here, we report that the bumblebee, *Bombus terrestris*, uses path integration while walking over short distances in an indoor arena. They estimate accurate vector distances after displacement and orient by artificial celestial cues. Walking bumblebees also exhibited systematic search patterns when home vectors failed to lead them accurately back to the nest, closely resembling searches performed by other species under natural conditions. We thus provide a robust experimental system to test navigation behavior in the laboratory that reflects most aspects of natural path integration. Importantly, we established this assay in an animal that is both readily available and resilient to invasive manipulations, as we demonstrate with the retention of the homing behavior post-anesthesia and surgery. In the future, our behavioral assay can therefore be combined with current electrophysiological techniques, opening a path toward directly probing the neural basis of the sophisticated vector navigation abilities of bees.

INTRODUCTION

Path integration is a computational strategy that allows animals to maintain an internal estimate of their position within the environment with respect to a point of origin. Many species use path integration for navigation, particularly when they return to a location of interest after traveling away from it. During this process, an animal monitors its angular and translational displacements to continually compute the distance and angle to the origin of its excursion, representing the most direct path back to the reference point, the home vector. This home vector can guide steering movements to enable the animal to return to its original location.¹

The use of path integration has been most clearly demonstrated in central place foraging animals that return to a home location between foraging bouts,^{2–8} and has been most thoroughly studied in hymenopterans, including desert ants⁹ and honeybees.¹⁰ These champion navigators are able to accurately locate their nests/hives after hundreds of meters of tortuous travel. To update their positional estimate, hymenopteran insects primarily rely on external (allothetic) cues as sources for directional information and self-generated movement (idiothetic) cues for distance information. Although many cues can be used to obtain directional information, celestial cues are among the most prevalent. Many species, including hymenopterans,

primarily orient using the skylight polarization pattern^{5,10–15} and the solar azimuth.^{5,16–20} For measuring distance, honeybees use optic flow,^{21,22} whereas desert ants primarily use a stride integrator.^{23,24}

Although path integration is a robust navigation strategy, positional estimates obtained using idiothetic information accumulate error and, therefore, are not perfect. Responding to this error, path-integrating animals initiate a stereotyped search behavior if they fail to reach their goal after traveling the distance indicated by their home vectors. These patterns are generally composed of loops that are centered around the initiation point of the search and increase in size over time, a highly effective strategy for locating a goal in the absence of a positional marker.^{3,4,6,25–27}

Like honeybees, bumblebees (*Bombus terrestris*) forage at flower patches for pollen and nectar before returning to their nests. This species has become an important model in behavioral neuroscience^{28–30} due to its relatively small, manageable hive size, commercial availability, large body size, and robustness to invasive experimental procedures. All these characteristics are key assets to experimental investigations in the laboratory, including those aimed at unraveling the still enigmatic neural basis of path integration. However, although bumblebees are closely related to honeybees and therefore likely use path integration-based vectors during navigation as well, this ability



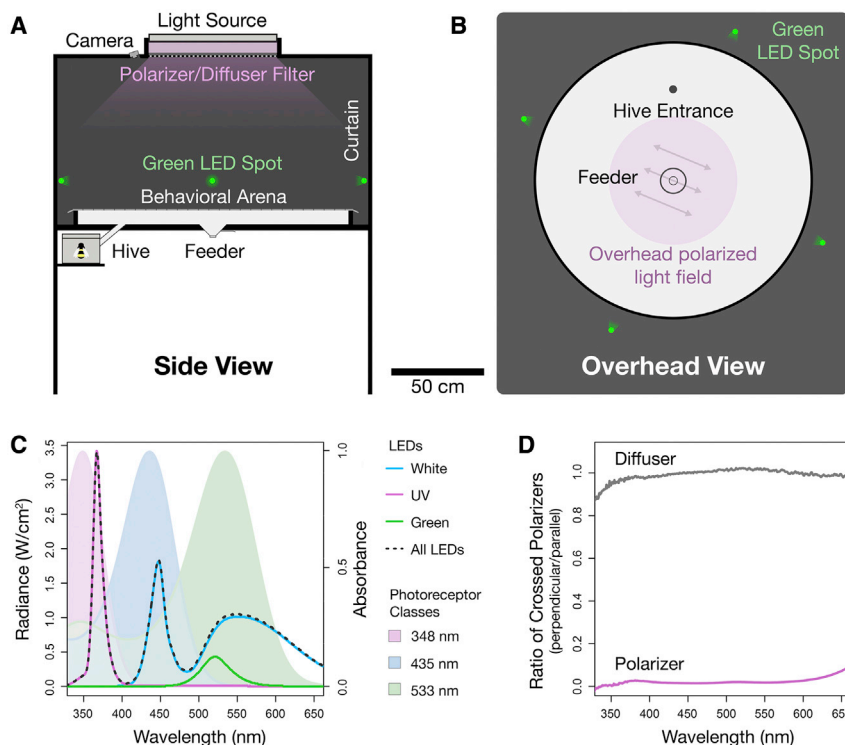


Figure 1. Navigation arena design

(A) Side view.

(B) Overhead view. The circular arena (150 cm diameter) was directly connected to a hive via an entrance located 25 cm from the arena's periphery (filled circle). Sugar water and pollen were provided in a conical feeder at the arena center. Neither the feeder nor hive entrance were visible to bees in the arena. The arena was illuminated by an overhead light source that could either be highly polarized or depolarized. Four green point sources were positioned at 90° intervals around the arena. Experiments were video recorded from above.

(C) Irradiance spectra of light available at the center of the arena (dashed line). Violet and blue curves: irradiance of the isolated UV and white LEDs from the overhead light source. Green curve: irradiance of the green LED point source (at a distance of 20 cm). Curves with colored area under the curve: Absorbance spectra of photoreceptor classes in the eyes of *Bombus terrestris dalmatinus* (from Skorupski et al.³⁴).

(D) Ratio of light transmitted through the overhead filter and a second crossed polarizer (perpendicular orientation/parallel orientation) when either the polarizer side (violet) or diffuser side (gray) faced toward the arena.

See also Figure S1.

has to date not been unambiguously demonstrated in this species.

Recent gains in understanding the neural circuitry that likely underlies path integration in bees has yielded a region of the arthropod brain termed the central complex as the main cellular substrate for navigation control. This progress is based on detailed anatomical descriptions of insect central complexes^{31,32} combined with physiological data and anatomically constrained computational models for path integration in bees.³³ However, to link neural circuits of the central complex to the sophisticated path integration abilities of hymenopteran insects, circuit data need to be combined with functional and behavioral data in a single model species. Despite the lack of behavioral data but given their robustness for electrophysiology and our advanced understanding of their central complex circuits, we propose bumblebees as the prime model for this endeavor.

Here, we developed a robust behavioral assay demonstrating vector navigation in bumblebees. By exploiting walking behavior on a flat surface over relatively short distances, using artificial skylight cues as sole sources of directional information, our assay allows examination of hymenopteran vector navigation systematically in a controlled setting. Importantly, we show that home vector memories are retained post-anesthesia and surgery, demonstrating that the assay is also highly amenable to electrophysiological experiments and other invasive approaches. We thus not only reveal robust path integration abilities in walking bumblebees but also offer the chance to further dissect complex vector navigation at the behavioral as well as neural level within a single insect species.

RESULTS

Bumblebees maintain an active hive while walking in the laboratory

First, we asked whether a bumblebee hive consisting entirely of walking bees would maintain a steady population of foraging individuals that can be probed for navigational abilities. We thus allowed a hive of bumblebees with clipped wings to forage in a featureless circular arena 1.5 m in diameter. The arena and a hive placed under it were joined by a tube so that the entrance was hidden from view while the bees were foraging. Equally hidden from view was a conical feeder, which was filled with pollen and sugar solution and placed at the center of the arena (Figures 1 and S1). Illumination was provided by an overhead light source consisting of white and ultraviolet LEDs and a composite filter constructed of a polarizer and a diffuser (Figures 1A and 1C). This light source could either deliver a linearly polarized light field, as was the case during the initial observations of the hive, or a depolarized light field (Figure 1D). Foraging paths to and from the feeder were video recorded from above (Figure 1).

Over a period of 22 days, initial observations showed that the hive's foragers were actively making walking foraging trips throughout the entire 12-h light period of the day (Figure 2A). Peak activity periods of individuals occurred at different times of the day from one another (Figure 2B), and the aging hive actively produce cohorts of new bees who successively took over the foraging duties of older bees (Figure 2C). Thus, despite the unnatural state of the hive and the artificial environment, our hives of walking bees appeared healthy, and the collective activity of the hive largely resembled that reported for hives of flying bees placed in large greenhouses.³⁵

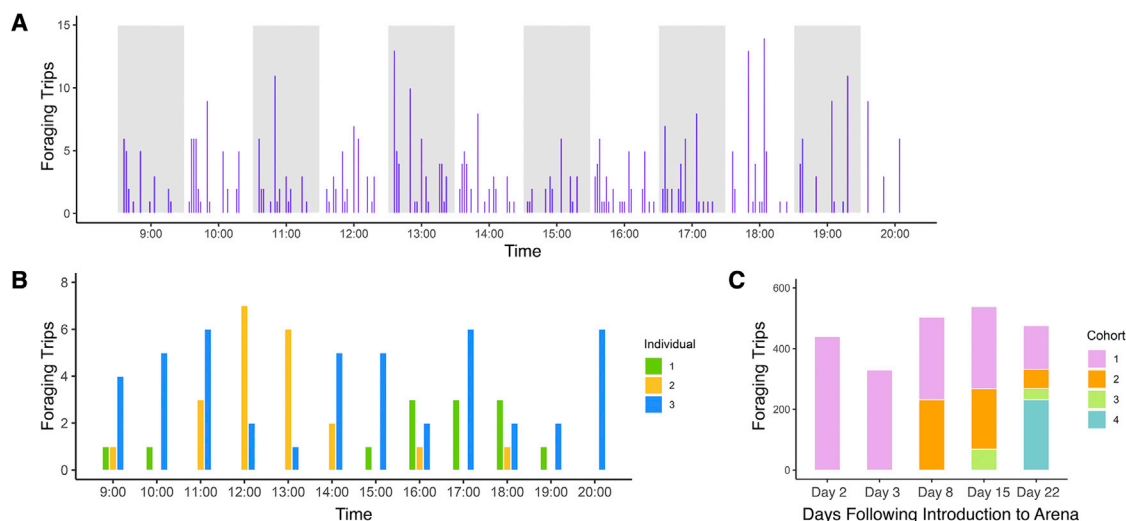


Figure 2. Foraging activity of a hive of walking *Bombus terrestris* in the arena

(A) Activity over the course of one representative day light period (day 2 after introduction to the arena); bees were active throughout the entire light period (9:00–21:00). Alternating gray and white columns: hourly intervals. Violet bars: number of times a single individual made a foraging trip from the hive to the feeder and back again over the course of the corresponding 1-h interval.

(B) Individuals are active at different times of the day. Colored bars: number of times a single individual made a foraging trip from the hive to the feeder and back again over the course of the corresponding 1-h interval.

(C) The hive actively produces new individuals that replace older foragers. Colored bars: number of foraging trips made by individuals of each cohort per day. Individuals that were new during an observation day were assigned to a new cohort.

Bumblebees navigate using path integration over short distances while walking

Next, we asked whether walking bumblebees would use path integration to return home after foraging at the feeder. When naive bees entered the arena for the first time, we observed examples of straight homing paths after bees had explored the arena, consistent with path integration (Figure S2). However, these initial trips were often exploratory in nature across the population of bees in the arena. We therefore allowed bees to familiarize themselves with the arena to observe consistent homing behaviors. After 3 days of acclimation to the arena with the overhead polarization field present, bees were released individually from the hive and allowed to locate the feeder. After feeding, bees executed a generally well-directed homeward path, either toward or opposite to the hive's location (Figure 3B; Video S1). This bimodal orientation is expected since a polarization pattern used as a sole orientation cue indicates only an axis of orientation rather than an absolute direction. If the hive was not encountered at the end of the homeward path, the bees initiated a search behavior, suggesting that *B. terrestris* may use path integration to locate its hive while walking over short distances.

To conclusively demonstrate that *B. terrestris* were indeed homing using a vector-based mechanism, we observed homeward paths of foraging animals after passively displacing them to a second arena. Bees were individually allowed to exit the hive and locate the feeder, where they were trapped. The closed feeder containing the bee was moved to a second, identical arena that lacked the hive entrance. The bees were either displaced to the center of the second arena (the same position it would be in the first arena) or to a position near the edge of the second arena. For edge displacements, a path integration-based home vector should direct the bees to the fictive hive's

position at the arena center, coinciding with the worst potential location of the hive entrance if directed by an image-matching strategy. Therefore, the lengths of homeward paths from both center- and edge-displacement experiments should be equal, both representing the length of the home vector (Figure 3A). Conversely, if the bees navigate to the position corresponding to that of the actual hive in the first arena after edge displacements, they would have to be locating their hives by an alternate strategy, for instance by visually matching their position to a unique visual environment at the hive's location.

In all experiments, the bees' behavior was consistent with a path integration-based strategy. When they were displaced to the second arena, homeward paths lengths from both edge- and center-displacement experiments were close to the beeline distance from the hive to the feeder in the first arena (center, $90.6\% \pm 6.4\%$ of beeline distance; edge, $104.2\% \pm 13.1\%$ of beeline distance [mean $\pm$ SD]; Figures 3C and 3D). As predicted for the edge-displacement experiments, the endpoint of the home vector (indicated by the initiation of search behavior) was located close to the center of the second arena (Figure 3C, center; Video S2).

When one of the two potential displacement locations were not in use, a fitted plug sealed the hole in the arena where the feeder could be fit. In order to control for any effect the small crack between the plug and the base of the arena may have had on influencing a homing bee to end its home vector and start a search pattern, edge-displacement experiments were repeated when the orientation of the polarized light field over the second arena was offset 45° from the polarized light field in the first arena, orienting homing bees away from the center of the arena and the plug that was located there (for detailed analysis of polarization orientation, see results below). In this condition,

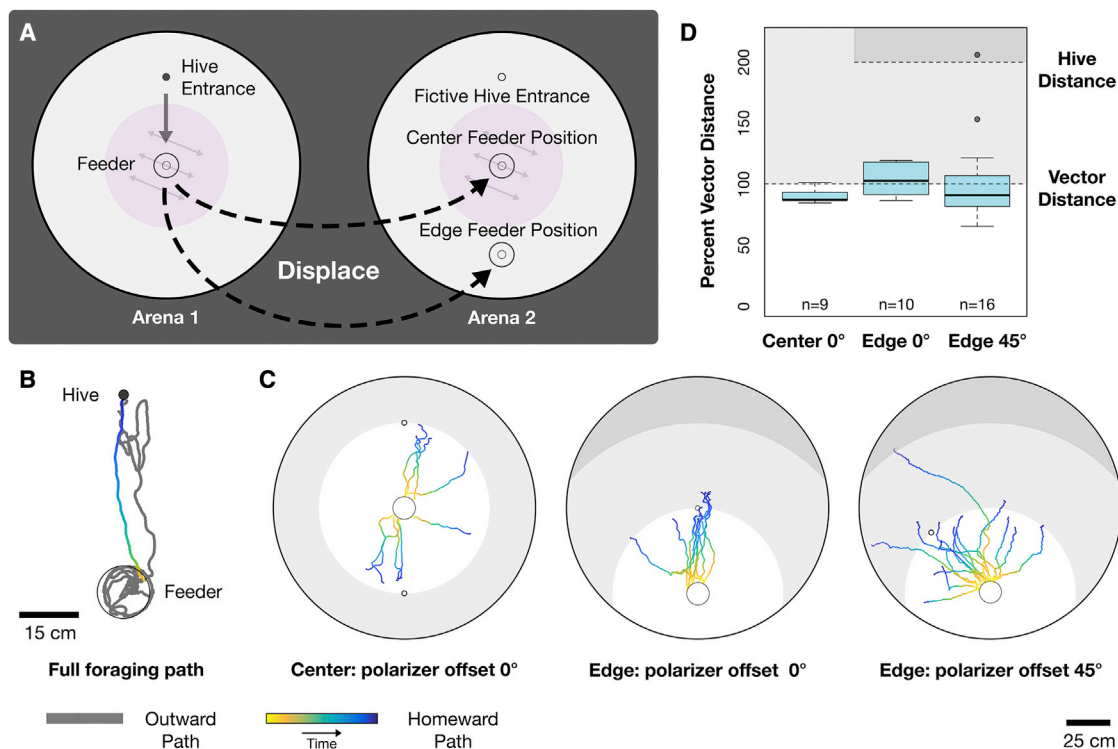


Figure 3. Walking *Bombus terrestris* use path integration to navigate back to the hive while foraging

(A) Experimental design: after finding the feeder, bees were trapped and displaced (dashed arrows) to the center or the edge of a second, identical arena that lacked the hive entrance. If the bees use path integration while homing in the arena, they should, after displacement, travel a length equal to the distance between the feeder and the hive entrance in the first arena (vector distance).

(B) Example of a foraging path from and to the hive when an animal was not displaced.

(C) Path tracings from all homeward paths before search behaviors were initiated during the three displacement conditions. Shades of gray: regions of the arena further away from the release point than the vector distance (light gray) and further than the hive distance (dark gray, only for edge displacements). For center displacements, hive distance and vector distance coincide.

(D) The percentage of the vector distance (feeder to hive in arena 1) traveled during homeward paths in arena 2 for all three displacement conditions. Bars, medians; boxes, lower and upper quartiles; whiskers, sample minima and maxima; points, outliers; horizontal-dashed lines, vector distance (all conditions) and the distance between displacement location and fictive hive entrance (edge displacements). Shaded regions as in (C).

See also [Figure S2](#) and [Table S3](#).

most bees followed the rotation of the polarizer and produced home vectors similar in length to the beeline distance from the hive to the feeder in the first arena ($100.7\% \pm 35.5\%$ of beeline distance [mean $\pm$ SD]; [Figures 3C and 3D](#)). Taken together, the results from our displacement experiments strongly support the hypothesis that foraging bumblebees use path integration while walking.

Although path integration most easily explains the consistent lengths of the homeward paths, it cannot explain all directional choices the bees made. We noted that bees did not orient toward the wall of the arena during the first edge-displacement experiment so that the expected bimodal orientation pattern became unimodally directed toward the fictive nest. Similarly, during edge-displacement experiments with a 45° polarizer offset, some bees were biased away from the arena wall closest to where the fictive nest should have been located according to a path integration home vector in that situation. These observations indicate that the proximity to the walls (and likely the visual stimuli they offer) additionally influences the homeward navigation of bees in the arena. Nonetheless, following path

integration-based vector information is clearly the dominant navigation strategy bees utilized during our experiments.

Walking bumblebees use simulated celestial cues for orientation

Path integration requires that an animal possess a mechanism to determine its orientation and an odometer to measure the distances it travels. Since *B. terrestris* were able to orient correctly toward their hive during homing, we next tested if the bees were using the orientation cues we provided in the arena to choose their homing directions. In two distinct experiments, these cues included either the zenithal polarization field, as was present in the earlier experiments, or a bright point source. A high-powered green LED served as the point source and was located at an azimuth of 157.5° from the hive entrance with an elevation of 21.8° when viewed from the center of the arena. In both situations, the tested cue was the sole light source in the arena.

Before both sets of experiments, bees were allowed to freely forage under the respective stimuli for 3 days. Following familiarization, once a bee located the central feeder, it was trapped and

the directional cue was rotated either by 90° from its original position (polarization field) or by 180° (green LED). If bumblebees use these cues to inform their orientation choices, their homeward paths should reflect the rotations of the stimuli (Figures 4A and 4D). This would result in a perpendicular orientation with respect to the hive for the polarized light field and in an orientation opposite of the hive for the green LED. In control trials, stimuli remained in the same position between outbound and homeward paths. For all trials with the polarized light field, path orientations were expected to follow a bimodal distribution, since an isolated polarized light pattern only offers an axis of orientation.

When the polarized field was static, homeward paths were oriented parallel to the direction of the hive (data were doubled to create a unimodal distribution; STAR Methods; $348.05^\circ \pm 40.2^\circ$, $p < 0.001$; Figure 4B; all orientation statistics are summarized in Tables S1 and S2). When the polarized field was rotated 90°, individuals oriented their homeward paths perpendicular to the direction of the hive (doubled data, $161.75^\circ \pm 32.6^\circ$, $p < 0.001$; Figure 4B; Video S3). To ensure bees were indeed orienting to the polarization pattern the filter created rather than other unintended artifacts of the filter, such as potential small differences in brightness across the filter, control experiments were conducted during which the filter was configured to provide a depolarized light field. Under this condition, bees no longer exhibited straight homeward paths, and these paths exhibited no significant orientation ($p = 0.526$; Figure 4C; Table S3). Taken together, these results show that *B. terrestris* use overhead polarization information when available to orient while walking.

During experiments when the position of the LED was flipped 180°, homeward paths were primarily oriented in the opposite direction of the hive ($197.45^\circ \pm 34.15^\circ$, $p = 0.001$). During control trials when the LED was left in place, homeward paths were oriented toward the hive ($17.36^\circ \pm 14.55^\circ$, $p < 0.001$; Figure 4E; Video S4). Since the homeward directions from the first set of these point source experiments appeared clustered directly opposite of the green LED rather than around the expected position of the hive, we examined whether this bias was consistent in a repetition of the experiment with independently trained bees. Orientations from the second set of trials were more clearly centered toward the direction of the hive in the predicted manner rather than away from the direction of the LED (fixed, $356.83^\circ \pm 20.43^\circ$, $p < 0.001$; rotated, $191.38^\circ \pm 26.07^\circ$, $p < 0.001$; Figure 4E; Table S4), suggesting that the bees perform true menotactic orientation rather than simple negative phototaxis. Overall, these results indicate that *B. terrestris* can orient using a bright point source while walking. During these experiments, the bees may have interpreted the bright LED as the Sun, although it is also possible that the light was interpreted as an Earth-bound cue.

Bumblebees prefer to orient using a point source rather than overhead polarization patterns

Similar to our bumblebees, many animals can use more than a single cue for orientation. When orientation responses are examined in a natural environment or in a laboratory setting, some cues are weighted more heavily than others and therefore dominate in experimentally generated cue-conflict situations.^{5,10,12,18,36,37} Since we demonstrated that *B. terrestris* will orient to overhead polarization patterns and a point source when both cues were

in isolation, we next were interested in which of the two cues the bumblebees preferred to orient with when both were available in concert.

As in previous experiments, bumblebees were allowed to freely forage in the arena for 3 days while the overhead polarization field and the point source were both present in the arena. These cues were arranged so that the bright LED was located 157.5° from the hive and 90° with respect to the angle of polarized light at an elevation of 21.8° , loosely mimicking the natural skylight situation close to dusk or dawn. After familiarization, bees were allowed to forage individually. Once a bee was trapped in the feeder, both orientation cues were rotated by 90° . Considering the results from the previous experiments, bees were predicted to orient 90° from the direction of the hive in this condition. Finally, each orientation cue was individually rotated by 90° , whereas the other cue remained in its original position. If *B. terrestris* weights one of the two cues more strongly, the bees should follow the stimulus rotation only when the dominant cue had been rotated (Figure 5A).

When neither cue was manipulated, bees oriented significantly in the axis of the hive (doubled data, $7.39^\circ \pm 38.96^\circ$, $p < 0.001$). During experiments in which both cues were rotated, bees oriented perpendicular to the hive direction, as expected (doubled data, $181.52^\circ \pm 55.41^\circ$, $p = 0.016$). During trials when the orientation cues were rotated in isolation, bees ignored the rotation and oriented toward the hive when only the polarization pattern was rotated (doubled data, $349.8^\circ \pm 20.63^\circ$, $p < 0.001$) and followed the stimulus manipulation when only the point source was moved (doubled data, $169.31^\circ \pm 44.69^\circ$, $p = 0.004$). These results demonstrate that bees weighted the point source more highly than the polarization field in the navigation arena and consequently ignored the polarization stimulus when choosing a walking direction if it conflicted with the azimuth information of the point source (Figure 5B).

Bumblebees perform systematic search patterns when home vectors fail

In the previous displacement experiments, the end of a home run of a bee was defined by a sudden change in extended walking direction, revealing the bee's estimate of where its hive should be. At this point, the bees executed stereotyped search behaviors to locate their home, a key feature of path integration in many species.^{3,4,6,25,27} In most experiments, these searches were located around a nest position that was close to one of the arena walls so that the resulting search pattern was geometrically biased by the limits of the arena. We therefore analyzed search behaviors only from the edge-displacement experiments under polarized light fields oriented identically between the two arenas. In those conditions, searches started near the center of the arena and therefore were geometrically unbiased. These observed search behaviors were composed of loops that started and ended near the search initiation point, therefore being most focused around the search origin, and increased in size over time (Figures 6A and 6B).

In other path-integrating animals, homing usually consists of a straight, quick home vector followed by a contorted and often slower search pattern.^{3,27,38} In contrast, we found that although bumblebees exhibited search patterns that were much more tortuous than home vectors, walking speed remained relatively

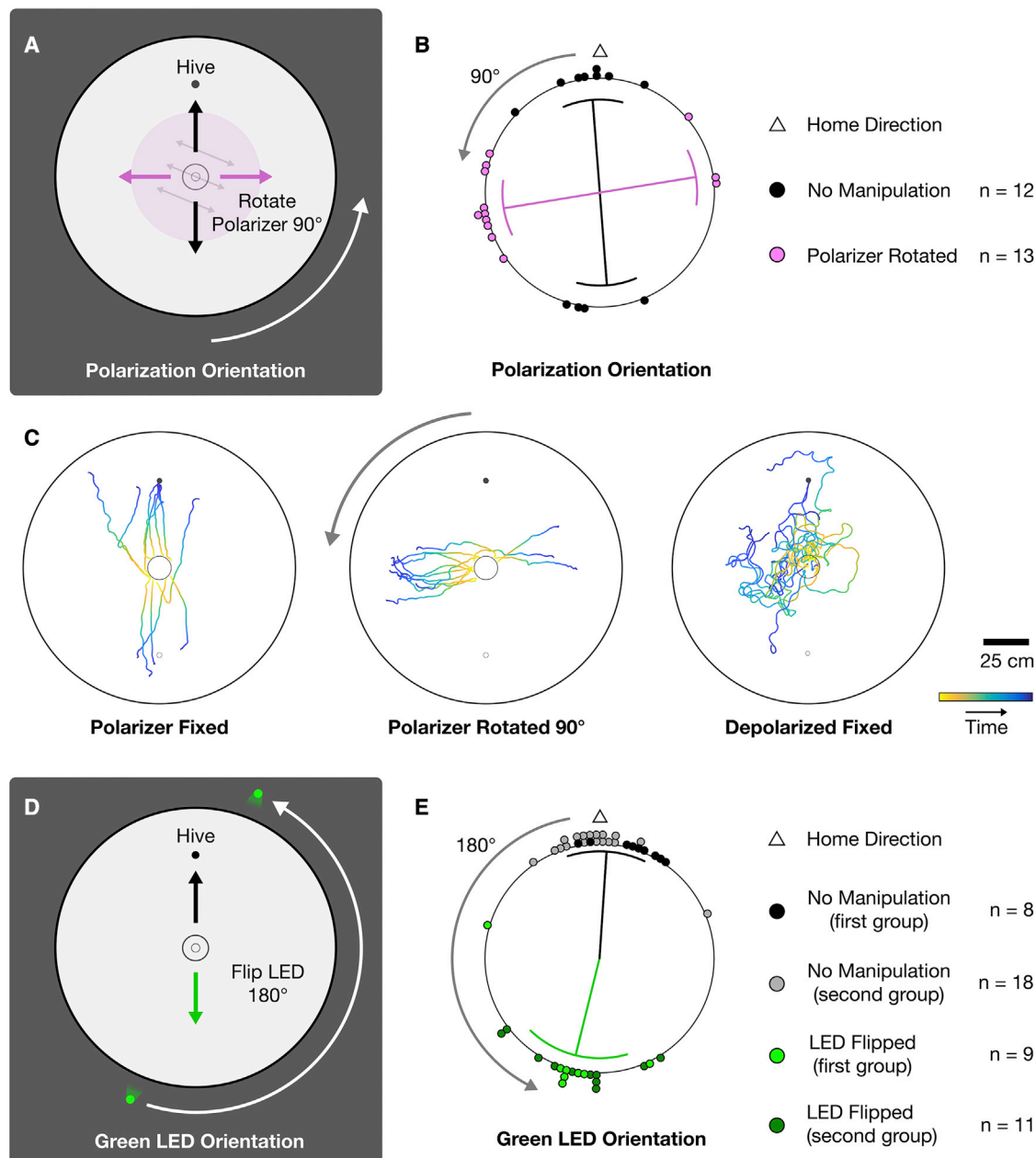


Figure 4. *Bombus terrestris* use directional cues present in the arena for orientation during path integration while walking

(A) Polarization orientation experimental design: An overhead polarized light field was rotated 90° from its original position once animals reached the feeder. Homeward paths were predicted to be oriented perpendicular to the direction of the hive if overhead polarization patterns were used for orientation.

(B) Orientations of homeward paths for polarization orientation experiments. The polarized light field was either fixed in place or rotated (direction: curved gray arrow). Arrows in each plot: mean vectors (arrow angles, circular mean; curved arrowheads, circular standard deviations; arrow lengths, strength of orientation $|\bar{r}|$). Both groups exhibited significant orientations and were significantly differently oriented from one another ($p < 0.05$).

(C) Path tracings of all home vectors before search behaviors were initiated under polarized and depolarized light. Straight homeward paths were only observed in 3 of 14 trials under depolarized light fields.

(D) Point source orientation experimental design: a green LED point source was flipped 180° from its original position once animals reached the feeder. Homeward paths were predicted to be oriented in the opposite direction of the hive if *B. terrestris* were orienting using the point source.

(E) Orientations of homeward paths for point source orientation experiments. All groups exhibited significant orientations, and manipulated groups were significantly differently oriented from not manipulated groups ($p < 0.05$). For summary of all orientation statistics, see [Tables S1](#) and [S2](#).

See also [Tables S3](#) and [S4](#).

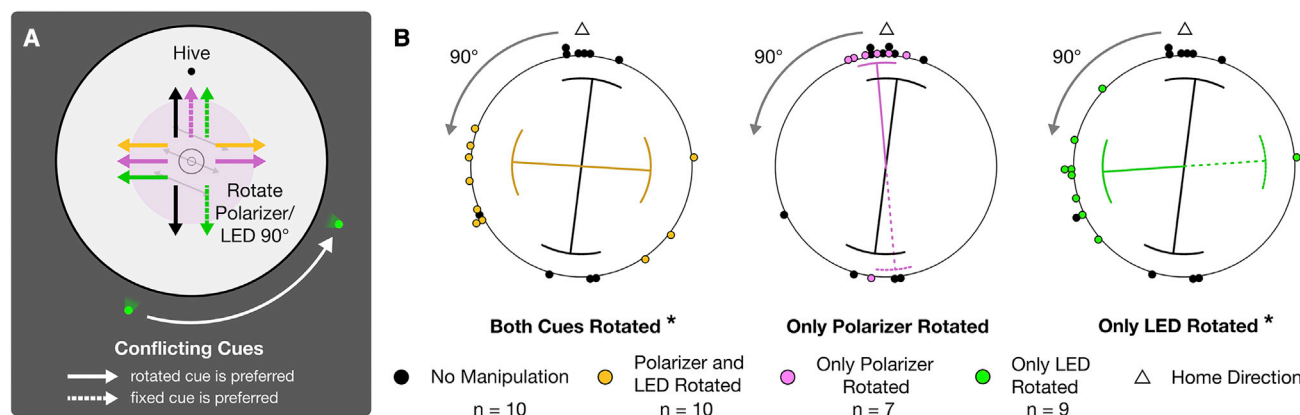


Figure 5. *Bombus terrestris* prefer to orient using a point source rather than an overhead polarization field in the arena while path integrating
(A) Conflicting cues experimental design: to determine if bumblebees preferred to use the polarized light field or the green LED for orientation when both are present in the arena, both cues were rotated 90° either together or in isolation. If the bees prefer one of the two orientation cues, they should only orient 90° from the hive when the dominant cue has been rotated.

(B) Orientations of homeward paths for conflicting cue orientation experiments. Under each condition, the orientation cues were either fixed in place or rotated (direction: curved gray arrows). All groups exhibited significant orientations ($p < 0.05$). Asterisks (*) indicate manipulation conditions with group orientations that were significantly differently oriented from trials when no manipulations occurred ($p < 0.01$). Arrows in each plot: mean vectors (arrow angles, circular mean; curved arrowheads, circular standard deviations; arrow lengths, strength of orientation $|\bar{r}|$). For summary of all orientation statistics, see [Tables S1](#) and [S2](#). See also [Tables S3](#) and [S4](#).

constant throughout both phases of homeward travel and over the duration of the search ([Figures 6D–6F](#)).

To examine whether walking bumblebee search patterns quantitatively resemble those observed in other species as well as those predicted by optimal search theory, we determined the rate of the bees' search expansion by measuring the farthest distance between the bee and the search initiation point during each successive increasing loop in the search, until bees reached a distance of 25 cm to the arena wall (the distance of the fictive hive entrance from the arena wall). This limit was imposed to prevent the constraints of the arena wall to impact the free expansion of the search. Assuming constant detection width of the nest, constant speed, and a Gaussian uncertainty distribution, optimal search theory predicts a parabolic shape of the search distribution to maximize search success. This results in an expected increase of search radius proportional to the square root of the search time, $\text{radius}_{\text{max}} \propto \text{time}^{0.5}$.¹ Data from desert ant searches²⁵ match this prediction, with the maximum search radius being proportional to $\text{time}^{0.48}$.¹ Similarly, the radii of searches in mantis shrimp following path integration home vectors expand at a rate proportional to $\text{time}^{0.43}$.²⁷ By fitting a power function to our data, we observed that the search patterns of walking bumblebees expand at a rate proportional to $\text{time}^{0.40}$ ([Figures 6C, 6G, 6H, and S3](#)). When only the initial expansion of the search is observed, further limiting the potential constraining effect of the arena wall, the expansion rate of the searches was proportional to $\text{time}^{0.47}$ ([Figure 6I](#)), very closely approximating the expansion of searches predicted by optimal search theory and matching those exhibited by other path-integrating animals.

Vector navigation behavior is resilient to anesthetization and surgery

Our assay offers a robust tool to dissect vector navigation in the laboratory over relatively short distances, an asset for future

electrophysiological investigations aimed at probing the neural basis of path integration. To confirm the utility of our assay for electrophysiology, we investigated the resilience of the bees' vector memories to invasive experimental procedures. This included subjecting the bumblebees to anesthesia and to a surgery similar to that required by electrophysiological studies targeting insect central brain regions.^{33,39–42}

Following familiarization under a polarized light field, once bees located the central feeder, they were trapped, removed from the arena, and anesthetized in a chill-coma. In a subset of bees, the head cuticle was cut with a sharp razor in three cuts over the central brain ([Figure 7A](#)), and the incisions were sealed with dental resin. Bees subjected to surgery were allowed 6 h of recovery time before experimentation, whereas bees that were only anesthetized were used immediately. In a third group of bees, the stability of the vector memory over time was analyzed by keeping the bees trapped in the feeder for 12 h before experimentation, without anesthetization. All bees were reintroduced to the center of the behavioral arena with the overhead polarized filter oriented either the same or orthogonal to its original position ([Figure 7B](#)).

Bumblebees that were returned immediately to the arena after anesthetization exhibited extended periods of stillness and undirected walking in and around the feeder, usually within a 14 cm radius of the feeder and lasting on average approximately 30 min, before making paths considered homing runs by our criterion. A notable proportion of these bees did not make clear homing runs at all ([Table S3](#)). Although a large enough sample to claim significant orientation was only collected during trials when the polarized filter was rotated 90° (doubled data, $188.86^\circ \pm 33.63^\circ$, $p < 0.001$; [Figure 7E](#)), the eventual post-anesthesia home runs from both groups were generally oriented as predicted by the overhead polarized light field. This indicated that the vector memories were anesthesia resistant but that sufficiently long recovery periods were needed for successful homing behavior.

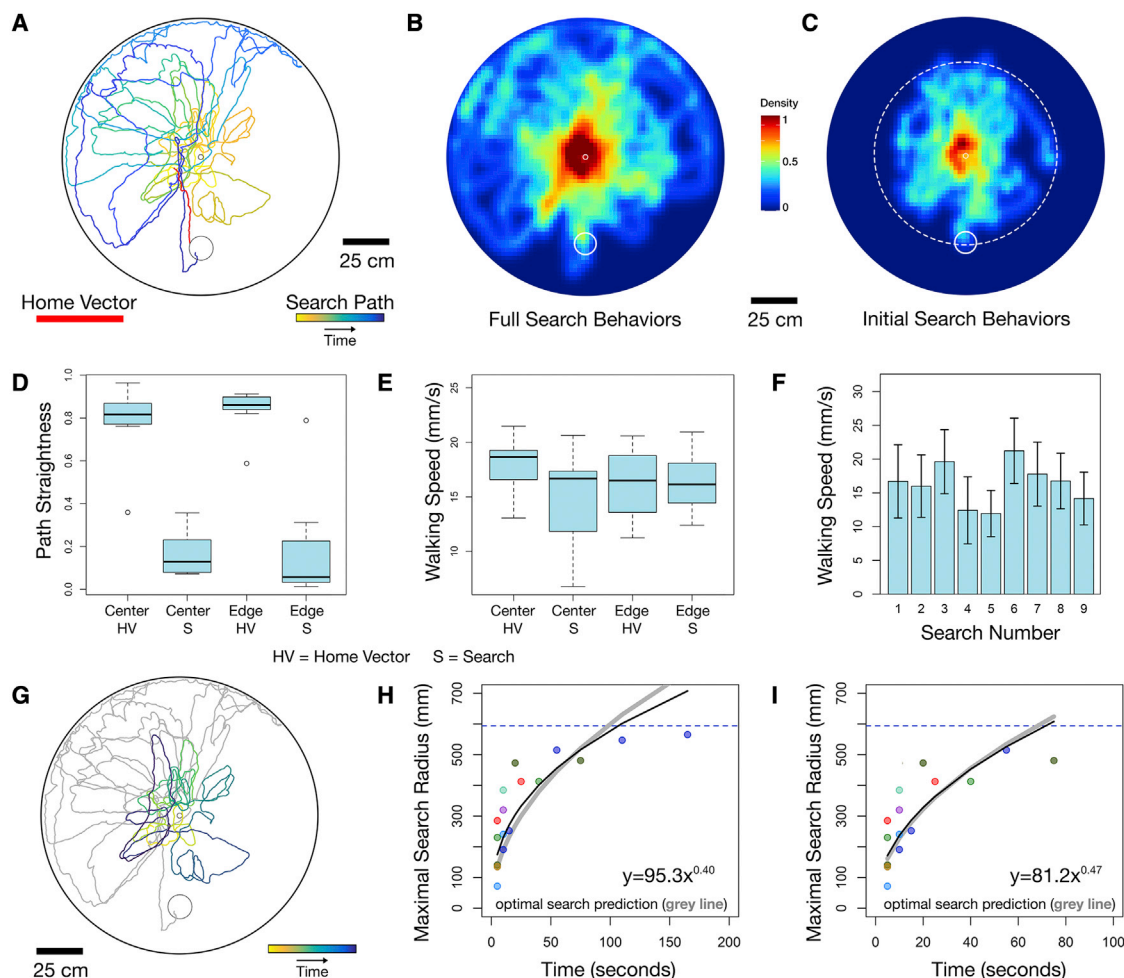


Figure 6. *Bombus terrestris* perform systematic search behaviors when path integration home vectors fail to locate the hive

(A) Example of a search behavior during a trial when a bee was displaced to the edge of arena 2. Larger empty circle, displaced feeder where bee was introduced; smaller empty circle, vector position of the fictive hive.

(B) A heat map of all search behaviors compiled during edge-displacement experiments with no polarizer offset.

(C) A heat map of all search behaviors in (B) up to a distance of 25 cm from the arena wall (dashed circle) used for analysis in (H).

(D) Straightness of paths during the homing and search phases of center- and edge-displacement experiments (where path straightness = beeline distance/path length). Home vector paths were straighter than search paths ($p < 0.001$; all other pairwise comparisons, $p > 0.1$).

(E) Walking speed remained approximately constant during homing and search phases ($p = 0.512$; $\chi^2 = 3.284$). Bars, medians; boxes, lower and upper quartiles; whiskers, sample minima and maxima; points, outliers.

(F) Mean walking speed of each bumblebee during the search phase of edge-displacement experiments. Bars show average walking speeds over 1-s intervals throughout a search for each bee, with error bars ($\pm$ SD) indicating relatively constant walking speeds during searches.

(G) Search behavior from (A) but with the colored section highlighting completed loops before the search expanded to a distance of 25 cm to the wall, as used for analyses in (H).

(H) Absolute radii of searches in (C) with at least one completed loop before the search expanded to a distance of 25 cm from the arena wall ($n = 8$). The maximum distance of each increasing search loop from the search origin is plotted against time after search initiation. Colors, individual searches; black line, the power function of best fit for all data, yielding a search expansion relation with $\text{radius}_{\text{max}} \propto \text{time}^{0.4}$. Optimal search theory predicts that searches should expand by $\text{radius}_{\text{max}} \propto \text{time}^{0.5}$ (gray line). Dashed blue line: the distance between the search origin and the 25 cm limit from the arena wall for the search plotted in the same color.

(I) Same as (H) except constrained to the first 100 s of search. The initial search expansion ($\text{radius}_{\text{max}} \propto \text{time}^{0.47}$) better fits the expansion predicted by optimal search theory.

See also Figure S3.

Confirming the high stability of the observed vector memories, bees that were caught and trapped at the feeder for 12 h exhibited homing behaviors comparable with those released immediately in the arena (doubled data; fixed, $357.49^\circ \pm 54.89^\circ$; rotated 90° , $200.32^\circ \pm 51.45^\circ$, both $p < 0.01$; Figure 7F).

Thus, extended recovery times post-anesthesia of at least up to 12 h are possible without significant detriment to the homing behaviors we present.

Finally, bees were tested after anesthetization, surgery, and 6 h of recovery time. When the polarized field was oriented the

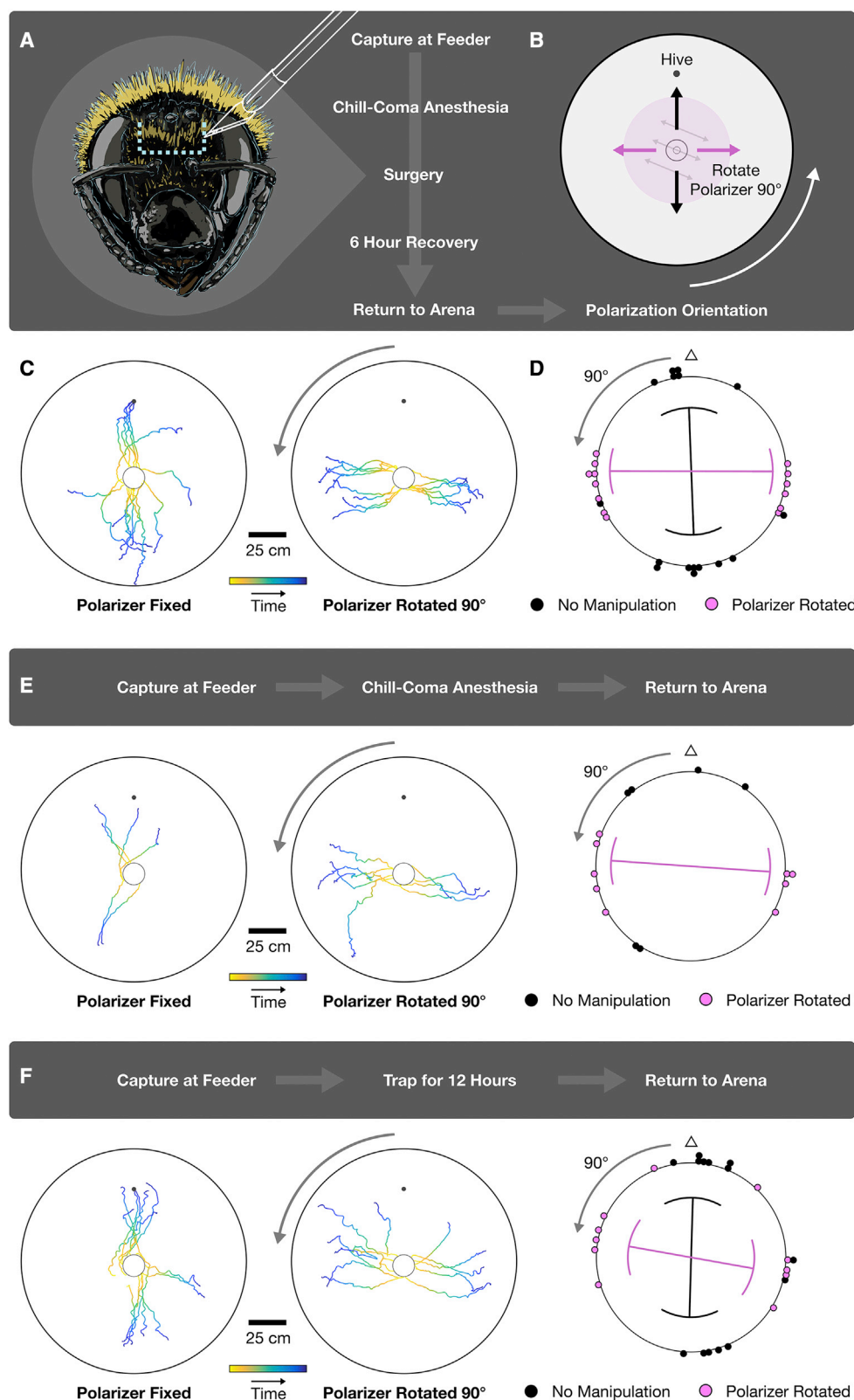


Figure 7. Vector navigation in *Bombus terrestris* is resilient to anesthetization and surgery

(A and B) Surgery experimental design (A). During surgery, head cuticles were cut along the dotted lines and subsequently patched. After recovery, bees were returned to the arena and polarization orientation experiments were performed as in (B).

(legend continued on next page)

same as during the bees' outward paths, homeward paths were oriented parallel to the direction of the hive (doubled data, $351.46^\circ \pm 51.11^\circ$, $p < 0.001$; Figures 7C and 7D) and were oriented perpendicular to the direction of the hive when the polarized field was rotated 90° (doubled data, $180.51^\circ \pm 31.52^\circ$, $p < 0.001$; Figures 7C and 7D). These post-surgery orientations were not significantly different from the earlier orientations to the same polarized light field of bees not exposed to anesthetics and surgery (both polarizer fixed and rotated 90° , $p > 0.1$). The extensive still and meandering behaviors of bees released immediately in the arena post-anesthetization were not observed in the majority of trials including a 6-h recovery period, indicating that extended recovery times are required to limit negative effects of chill-coma anesthesia on behavioral performance.

In summary, the homeward orientations of bees subjected to memory challenges consistently resembled that of untreated bees. Thus, the vector memories we examined in this study are resilient to the anesthesia and surgical methods required for physiological investigations of central brain regions.

DISCUSSION

Our results are the first to conclusively demonstrate that bumblebees use path integration to navigate and is one of the first to show that bees path integrate over short distances while walking. Previous work on navigation in walking honeybees⁴³ found that bees forced to walk performed waggle dances very similar to those of flying bees. These walking bees transitioned from round dances (indicating feeders at short distances) to waggle dances (indicating feeders at further distances) at a much earlier stage (3–4 m rather than 50–100 m). Thus, walking foraging resembled a scaled down version of foraging during flight in key aspects. Importantly, when honeybees were forced to approach and return from a feeder along a curved or 90° angled path, the waggle dances they performed back in the hive correctly indicated the beeline direction of the feeder, suggesting that the walking bees performed path integration during the outbound trip. Reflecting these results in walking honeybees, we found that in the laboratory, walking bumblebees path integrate by using polarized light fields and point sources as directional cues. They also estimate vector distances accurately; however, it remains unclear what mechanism they use to track distances in our arena. Stride integration, optic flow, or a combination of the two are likely contenders. Walking bumblebees indicate the arrival at their estimated hive position by performing stereotyped search patterns that resemble those found in other path-integrating species. Taken together, the ability to path integrate while walking appears to be a feature present across a wide range of hymenopterans, including species that usually fly.

Although our assay provides a novel and robust paradigm to dissect vector-based navigation, the question arises of how biologically relevant navigation abilities over short distances in walking bumblebees are, as these animals typically fly over long distances while foraging. Despite their aerial lifestyle, bumblebees almost exclusively walk in the hive and will also occasionally walk over short distances while away from the hive in nature.⁴⁴ Some species, as reported for an Amazonian bumblebee (*Bombus transversalis*), even build extensive walking trails around their nest and thus resort to a relatively terrestrial lifestyle.⁴⁵ Chittka et al.²⁸ found that bumblebees (*Bombus impatiens*) also choose to walk rather than fly when forced to forage in darkness. During this study, bumblebees could walk to a feeder but were unable to make oriented homeward paths after feeding, similar to our experiments under a depolarized light field. As suggested by modeling studies,^{1,46–48} path integration relying on purely idiothetic (self-motion) directional cues likely accumulates too much error to allow for successful homeward navigation in situations where clear environmental directional information is absent. In each case, bees must maintain a stable navigational sense while walking, in addition to their well-known ability to navigate while flying over long distances. Path integration offers a solution to that need and could be implemented by swapping the potential sources of distance information, e.g., between optic flow (used during flight) and a stride integrator (used during walking), thereby co-opting the same neural machinery for flight and walking.

Our data demonstrate that bumblebees orient with the directional cues we provided, yet the artificial nature of these cues limits generalizations to a natural environment and leaves some uncertainty about how the bees interpret them. As may be the case in our navigating bumblebees, honeybees interpret a long-wavelength light spot as the direction of the Sun⁴⁹ and can also use a strong artificial polarization field to orient their waggle dances, communicating their previous foraging paths away from the hive under the natural sky.¹⁰ However, animals might utilize cues in a different manner if they differ in key aspects from nature. For example, desert ants will exhibit phototaxis toward the Sun when its spectral composition is limited to long wavelengths, no longer recognizing it as an orientation cue,¹² and honeybees exhibit positive phototaxis after being kept in the dark.⁵⁰ Although only positive phototaxis of foraging hymenopterans has been reported, the possibility of our bees being negatively phototactic to the point source rather than using it as a dynamic orientation cue, although unlikely (Table S4), cannot be completely discounted.

Similarly, although our data provide a clear cue hierarchy for the specific stimulus conditions of our arena, orientation preferences may differ in nature. Both absolute and relative light intensities for natural celestial cues differ from the artificial cues in the

(C) Path tracings of home vectors before search behaviors were initiated.

(D) Orientations of homeward paths for surgery experiments. The polarized light field was either fixed in place or rotated (direction: curved gray arrow). Arrows in each plot: mean vectors (arrow angles, circular mean; curved arrowheads, circular standard deviations; arrow lengths, strength of orientation ($\bar{r}$)). Both groups exhibited significant orientations and were significantly differently oriented from one another ($p < 0.05$; fixed, $n = 16$; rotated, $n = 14$).

(E) Same as (A)–(D) for chill-coma anesthesia experiments. Only the polarizer rotated group exhibited a significant orientation ($p < 0.05$; fixed, $n = 6$; rotated, $n = 9$).

(F) Same as (A)–(D) for 12-h time-delay experiments. Both groups exhibited significant orientations and were significantly differently oriented from one another ($p < 0.05$; fixed, $n = 14$; rotated, $n = 11$). For summary of all orientation statistics, see Tables S1 and S2.

See also Table S3.

arena. Though orientation behaviors should be robust to such changes, given that these intensities vary highly in nature as well, cue preferences in bumblebees might change under more naturalistic conditions. Von Frisch¹⁰ found that when the Sun and a small overhead polarized light cue were placed in conflict, honeybees will orient their dances solely to the Sun. However, the influence of the polarizer became stronger with increased amounts of UV light transmitted. Celestial cue preferences in honeybees thus depend on the relative strengths of the two cues,⁵¹ a finding that is also true in ants³⁶ and dung beetles³⁷ and is likely to be true for orientation cue preferences in bumblebees as well.

Beyond demonstrating vector navigation abilities in walking bumblebees, our data also yielded initial insights into the nature of the vector memories bees employed in our assay. These memories of walking bumblebees were resilient to anesthesia and surgery and persisted over a 12-h period. As in previous studies of insect path integration following chill-coma anesthesia, the orientations of homeward paths following anesthesia were maintained. Although in ants⁵² and dung beetles⁵³ the distances of home vectors were diminished after cooling, these distances were approximately correct across all manipulations in our experiments, suggesting very stable vector memories, potentially due to a long-term storage mechanism of home vector memories in bumblebees. However, experiments with ants revealed that these insects are able to correctly home after anesthetization, but only if they are in a familiar environment containing landmarks.⁵² Unlike in our edge-displacement experiments, we cannot completely rule out that a similar image-matching strategy may have contributed to guiding the bees to the location of the hive after recovery from anesthetization in our study as well.

Interestingly, the orientation results of our surgery experiments with a 6-h delay between capture and release at the feeder revealed that our bees were not time compensating for any circadian shifts in the overhead polarization pattern. The natural skylight polarization pattern continuously shifts throughout the day, and some animals, like honeybees, have been shown to time compensate their orientation responses in anticipation of this celestial movement.¹⁷ After 6 h, the natural celestial polarization pattern would have rotated substantially. Since no shift in orientation was observed after the 6-h recovery time, our assay does not capture any time compensation that might occur under natural conditions. Either our bees adapted their time-compensation mechanism to the static polarization field during the familiarization days before experimentation or they interpreted the artificial polarization field as an orientation cue that is distinct from a truly celestial one.

Overall, the behavioral paradigm established here can be used to test many navigational related behaviors beyond what we have described, including mechanisms of odometry during walking. It also offers the chance to dissect more sophisticated strategies, including the generation of food vectors and novel shortcuts, landmark learning, integration of landmark and vector navigation, and developing optimal routes while foraging from multiple places (for hypotheses of how insects may neurally accomplish some of these tasks by using vector navigation, see Le Mo  l et al.⁵⁴). Further, our behavioral system can serve as a powerful comparative tool between animals that utilize

similar navigation strategies. For example, mantis shrimp foraging underwater have been shown to exhibit strikingly similar path integration behaviors in comparably sized arenas with what we present here in walking bumblebees.^{5,27} Comparative behavioral studies across a wide range of species promise insights into how similar navigational problems are solved by evolutionary disparate animals navigating in different media.

Finally, due to the resilience of our bumblebees' vector memories to surgeries required for electrophysiological investigations, our paradigm offers a clear path toward probing the neural basis of path integration during ongoing behavior. Over recent years, understanding of the neural basis of navigation behaviors in insects has increased substantially due to detailed functional studies and neuroanatomical descriptions of navigationally relevant brain regions. From these studies, the central complex, a conserved set of arthropod brain regions, has emerged as the likely navigational control center that both integrates navigationally relevant stimuli^{33,39–41,55–58} and directs steering movements during navigation.⁴² By combining our behavioral arena with electrophysiological techniques used to record from either freely behaving animals⁴² or by transferring our paradigm to a virtual reality setup for tethered behavior,^{38,41,55,57–59} hypotheses regarding the underlying neural mechanisms for path integration vector memory formation³³ can be directly tested.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
- METHOD DETAILS
 - Experimental Arenas
 - Spectrometry
 - Hive Observations
 - Experimental Procedures
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2022.05.010>.

ACKNOWLEDGMENTS

This research has received funding from the European Union's Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement no. 101027405 (to R.N.P.) and the European Research Council under grant agreement no. 714599 (to S.H.), as well as the Swedish Research Council (2018-04851, to S.H.). We thank Allen Cheung for his insight into optimal search theory and advice for our search analyses. We are also grateful to Johan B  ckman (LU Biology electronics workshop) for technical assistance in building the arena lighting system and Lars Fredricksson (LU Biology mechanical workshop) for help with planning and constructing the behavioral arena and Nils Sundqvist for additional assistance. We thank James Foster

for statistical advice. Finally, we thank members of the Heinze lab, Dacke Lab, and the rest of the Lund Vision Group for helpful discussions.

AUTHOR CONTRIBUTIONS

Conceptualization, R.N.P. and S.H.; methodology, R.N.P.; investigation, R.N.P. and J.K.; analysis, R.N.P. and J.K.; writing – original draft, R.N.P.; writing – review & editing, R.N.P. and S.H.; funding acquisition, R.N.P. and S.H.; guidance, S.H.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as a member of the LGBTQ+ community.

Received: March 2, 2022

Revised: April 4, 2022

Accepted: May 5, 2022

Published: May 30, 2022

REFERENCES

- Heinze, S., Narendra, A., and Cheung, A. (2018). Principles of insect path integration. *Curr. Biol.* 28, R1043–R1058.
- Beugnon, G., and Campan, R. (1989). Homing in the field cricket, *Gryllus campestris*. *J. Insect Behav.* 2, 187–198.
- Durier, V., and Rivault, C. (1999). Path integration in cockroach larvae, *Blattella germanica* (L.) (insect: Dictyoptera): direction and distance estimation. *Anim. Learn. Behav.* 27, 108–118.
- Zeil, J. (1998). Homing in fiddler crabs (*Uca lactea annulipes* and *Uca vomeris*: Ocypodidae). *J. Comp. Physiol. A* 183, 367–377.
- Patel, R.N., and Cronin, T.W. (2020). Mantis shrimp navigate home using celestial and idiothetic path integration. *Curr. Biol.* 30, 1981–1987. e3.
- Dacke, M., el Jundi, B., Gagnon, Y., Yilmaz, A., Byrne, M., and Baird, E. (2020). A dung beetle that path integrates without the use of landmarks. *Anim. Cogn.* 23, 1161–1175.
- Mittelstaedt, M.-L., and Mittelstaedt, H. (1980). Homing by path integration in a mammal. *Naturwissenschaften* 67, 566–567.
- Etienne, A.S., Maurer, R., Berlie, J., Reverdin, B., Rowe, T., Georgakopoulos, J., and Séguinot, V. (1998). Navigation through vector addition. *Nature* 396, 161–164.
- Müller, M., and Wehner, R. (1988). Path integration in desert ants, *Cataglyphis fortis*. *Proc. Natl. Acad. Sci. USA* 85, 5287–5290.
- von Frisch, K. (1969). *The Dance Language and Orientation of Bees* (Harvard University Press).
- von Frisch, K. (1949). Die polarisation des himmelslichtes als orientierender faktor bei den tänzern der bienen. *Experientia* 5, 142–148.
- Wehner, R., and Müller, M. (2006). The significance of direct sunlight and polarized skylight in the ant's celestial system of navigation. *Proc. Natl. Acad. Sci. USA* 103, 12575–12579.
- Goddard, S.M., and Forward, R.B. (1991). The role of the underwater polarized light pattern, in sun compass navigation of the grass shrimp, *Palaemonetes vulgaris*. *J. Comp. Physiol. A* 169, 479–491.
- Hawryshyn, C.W., and Bolger, A.E. (1990). Spatial orientation of trout to partially polarized light. *J. Comp. Physiol. A* 167, 691–697.
- Dacke, M., Nilsson, D.-E., Scholtz, C.H., Byrne, M.J., and Warrant, E.J. (2003). Insect orientation to polarized moonlight. *Nature* 424, 33.
- Délye, G. (1911). Observations et remarques critiques sur le mécanisme de l'orientation chez les fourmis. *Rev. Suisse Zool.* 19, 303–338.
- Lindauer, M. (1960). Time-compensated sun orientation in bees. *Cold Spring Harb. Symp. Quant. Biol.* 25, 371–377.
- Goddard, S.M., and Forward, R.B. (1989). The use of celestial cues in the offshore escape response of the shrimp, *Palaemonetes vulgaris*. *Mar. Freshw. Behav. Physiol.* 16, 11–18.
- Leis, J.M., and Carson-Ewart, B.M. (2003). Orientation of pelagic larvae of coral-reef fishes in the ocean. *Mar. Ecol. Prog. Ser.* 252, 239–253.
- Dacke, M., el Jundi, B., Smolka, J., Byrne, M., and Baird, E. (2014). The role of the sun in the celestial compass of dung beetles. *Phil. Trans. R. Soc. B* 369, 20130036.
- Srinivasan, M.V., Zhang, S.W., and Bidwell, N. (1997). Visually mediated odometry in honeybees. *J. Exp. Biol.* 200, 2513–2522.
- Esch, H.E., Zhang, S.W., Srinivasan, M.V., and Tautz, J. (2001). Honeybee dances communicate distances measured by optic flow. *Nature* 411, 581–583.
- Wittlinger, M., Wehner, R., and Wolf, H. (2006). The ant odometer: stepping on stilts and stumps. *Science* 312, 1965–1967.
- Pfeffer, S.E., and Wittlinger, M. (2016). Optic flow odometry operates independently of stride integration in carried ants. *Science* 353, 1155–1157.
- Wehner, R., and Srinivasan, M.V. (1981). Searching behaviour of desert ants, genus *Cataglyphis* (Formicidae, Hymenoptera). *J. Comp. Physiol.* 142, 315–338.
- Hoffmann, G. (1983). The search behavior of the desert isopod *Hemilepistus reaumuri* as compared with a systematic search. *Behav. Ecol. Sociobiol.* 13, 93–106.
- Patel, R.N., and Cronin, T.W. (2020). Path integration error and adaptable search behaviors in a mantis shrimp. *J. Exp. Biol.* 223, jeb224618.
- Chittka, L., Williams, N.M., Rasmussen, H., and Thomson, J.D. (1999). Navigation without vision: bumblebee orientation in complete darkness. *Proc. R. Soc. Lond. B* 266, 45–50.
- Jin, N., Landgraf, T., Klein, S., and Menzel, R. (2014). Walking bumblebees memorize panorama and local cues in a laboratory test of navigation. *Anim. Behav.* 97, 13–23.
- Solvi, C., Al-Khudhairy, S.G., and Chittka, L. (2020). Bumble bees display cross-modal object recognition between visual and tactile senses. *Science* 367, 910–912.
- Sayre, M.E., Templin, R., Chavez, J., Kempnaers, J., and Heinze, S. (2021). A projectome of the bumblebee central complex. *eLife* 10, e68911.
- Hulse, B.K., Haberkern, H., Franconville, R., Turner-Evans, D.B., Takemura, S.-Y., Wolff, T., Noorman, M., Dreher, M., Dan, C., Parekh, R., et al. (2021). A connectome of the *Drosophila* central complex reveals network motifs suitable for flexible navigation and context-dependent action selection. *eLife* 10, e66039.
- Stone, T., Webb, B., Adden, A., Weddig, N.B., Honkanen, A., Templin, R., Wcislo, W., Scimeca, L., Warrant, E., and Heinze, S. (2017). An anatomically constrained model for path integration in the bee brain. *Curr. Biol.* 27, 3069–3085. e11.
- Skorupski, P., Döring, T.F., and Chittka, L. (2007). Photoreceptor spectral sensitivity in island and mainland populations of the bumblebee, *Bombus terrestris*. *J. Comp. Physiol. A Neuroethol. Sens. Neural Behav. Physiol.* 193, 485–494.
- Morandin, L.A., Lavery, T.M., and Kevan, P.G. (2001). Bumble bee (Hymenoptera: Apidae) activity and loss in commercial tomato greenhouses. *J. Econ. Entomol.* 94, 462–467.
- Wystrach, A., Schwarz, S., Schultheiss, P., Baniel, A., and Cheng, K. (2014). Multiple sources of celestial compass information in the Central Australian desert ant *Melophorus bagoti*. *J. Comp. Physiol. A Neuroethol. Sens. Neural Behav. Physiol.* 200, 591–601.
- Khalidy, L., Foster, J.J., Yilmaz, A., Belušić, G., Gagnon, Y., Tocco, C., Byrne, M.J., and Dacke, M. (2022). The interplay of directional information provided by unpolarised and polarised light in the heading direction network of the diurnal dung beetle *Kheper lamarcki*. *J. Exp. Biol.* 225, jeb243734.

38. Dahmen, H., Wahl, V.L., Pfeffer, S.E., Mallot, H.A., and Wittlinger, M. (2017). Naturalistic path integration of *Cataglyphis* desert ants on an air-cushioned lightweight spherical treadmill. *J. Exp. Biol.* **220**, 634–644.
39. Heinze, S., and Homberg, U. (2007). Maplike representation of celestial e-vector orientations in the brain of an insect. *Science* **315**, 995–997.
40. el Jundi, B., Warrant, E.J., Byrne, M.J., Khaldy, L., Baird, E.J., Smolka, J., and Dacke, M. (2015). Neural coding underlying the cue preference for celestial orientation. *Proc. Natl. Acad. Sci. USA* **112**, 11395–11400.
41. Beetz, M.J., Kraus, C., Franzke, M., Dreyer, D., Strube-Bloss, M.F., Rössler, W., Warrant, E.J., Merlin, C., and el Jundi, B. (2022). Flight-induced compass representation in the monarch butterfly heading network. *Curr. Biol.* **32**, 338–349. e5.
42. Martin, J.P., Guo, P., Mu, L., Harley, C.M., Roy, R.E., Martin, J.P., Guo, P., Mu, L., Harley, C.M., and Ritzmann, R.E. (2015). Central-complex control of movement in the freely walking cockroach. *Curr. Biol.* **25**, 2795–2803.
43. Bissetzky, A.R. (1957). Die tänze der bienen nach einem fussweg zum futterplatz. *Z. Vergl. Physiol.* **40**, 264–288.
44. Heinrich, B., and Heinrich, M.J.E. (1983). Heterothermia in foraging workers and drones of the bumblebee *Bombus terrestris*. *Physiol. Zool.* **56**, 563–567.
45. Cameron, S.A., and Whitfield, J.B. (1996). Use of walking trails by bees. *Nature* **379**, 125–126.
46. Cheung, A., Zhang, S.W., Stricker, C., and Srinivasan, M.V.v. (2007). Animal navigation: the difficulty of moving in a straight line. *Biol. Cybern.* **97**, 47–61.
47. Cheung, A., and Vickerstaff, R. (2010). Finding the way with a noisy brain. *PLoS Comput. Biol.* **6**, e1000992.
48. Cheung, A. (2014). Animal path integration: a model of positional uncertainty along tortuous paths. *J. Theor. Biol.* **341**, 17–33.
49. Edrich, W., Neumeyer, C., and von Helversen, O. (1979). "Anti-sun orientation" of bees with regard to a field of ultraviolet light. *J. Comp. Physiol. A* **134**, 151–157.
50. Menzel, R., and Greggers, U. (1985). Natural phototaxis and its relationship to colour vision in honeybees. *J. Comp. Physiol.* **157**, 311–321.
51. Able, K.P. (1991). Common themes and variations in animal orientation systems. *Am. Zool.* **31**, 157–167.
52. Pisokas, I., Rössler, W., Webb, B., Zeil, J., and Narendra, A. (2022). Anesthesia disrupts distance, but not direction, of path integration memory. *Curr. Biol.* **32**, 445–452. e4.
53. Yilmaz, A., Gagnon, Y., Byrne, M., Baird, E., and Dacke, M. (2022). Cold-induced anesthesia impairs path integration memory in dung beetles. *Curr. Biol.* **32**, 438–444. e3.
54. Moël, F., le, Stone, T., Lihoreau, M., Wystrach, A., and Webb, B. (2019). The central complex as a potential substrate for vector based navigation. *Front. Psychol.* **10**, 690.
55. Seelig, J.D., and Jayaraman, V. (2015). Neural dynamics for landmark orientation and angular path integration. *Nature* **521**, 186–191.
56. Giraldo, Y.M., Leitch, K.J., Ros, I.G., Warren, T.L., Weir, P.T., and Dickinson, M.H. (2018). Sun navigation requires compass neurons in *Drosophila*. *Curr. Biol.* **28**, 2845–2852. e4.
57. Lyu, C., Abbott, L.F., and Maimon, G. (2022). Building an allocentric travelling direction signal via vector computation. *Nature* **601**, 92–97.
58. Lu, J., Behbahani, A.H., Hamburg, L., Westeinde, E.A., Dawson, P.M., Lyu, C., Maimon, G., Dickinson, M.H., Druckmann, S., and Wilson, R.I. (2022). Transforming representations of movement from body- to world-centric space. *Nature* **601**, 98–104.
59. Taylor, G.J., Paulk, A.C., Pearson, T.W.J., Moore, R.J.D., Stacey, J.A., Ball, D., van Swinderen, B., and Srinivasan, M.V.v. (2015). Insects modify their behaviour depending on the feedback sensor used when walking on a trackball in virtual reality. *J. Exp. Biol.* **218**, 3118–3127.
60. Heinze, S., el Jundi, B., Berg, B.G., Homberg, U., Menzel, R., Pfeiffer, K., Hensgen, R., Zittrell, F., Dacke, M., Warrant, E., et al. (2021). A unified platform to manage, share, and archive morphological and functional data in insect neuroscience. *eLife* **10**, e65376.
61. Batschelet, E. (1981). *Circular Statistics in Biology* (Academic Press).

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Displacement Experiments Data	https://insectbraindb.org	InsectBrainDatabase: https://hdl.handle.net/20.500.12158/EIN-0000210.1
Polarization Orientation Experiments Data	https://insectbraindb.org	InsectBrainDatabase: https://hdl.handle.net/20.500.12158/EIN-0000212.1
Point Source Orientation Experiments Data	https://insectbraindb.org	InsectBrainDatabase: https://hdl.handle.net/20.500.12158/EIN-0000213.1
Conflicting Cues Experiments Data	https://insectbraindb.org	InsectBrainDatabase: https://hdl.handle.net/20.500.12158/EIN-0000214.1
Behavior Resilience Experiments Data	https://insectbraindb.org	InsectBrainDatabase: https://hdl.handle.net/20.500.12158/EIN-0000229.1
Experimental models: Organisms/strains		
<i>Bombus terrestris</i>	Koppert Biological Systems	Natupol Smart
Software and algorithms		
ImageJ (Fiji)	NIH	Version 2.3.0
ivTrace	Bielefeld University	Version 2.23
Matlab	MathWorks	Version R2020a
R	R Core Development Team	Version 4.0.2
R package: 'autoimage'	French, J	Version 2.2.3
R package: 'circular'	Lund, U et al.	Version 0.4-94
R package: 'ggplot2'	Wickham et al.	Version 3.3.5
R package: 'plotrix'	Lemon, J et al.	Version 3.8-2
R package: 'shape'	Soetaert, K.	Version 1.4.6

RESOURCE AVAILABILITY

Lead contact

Further information or materials associated with this research will be made available upon reasonable request to the lead contact author, Rickesh Nitin Patel (e-mail: rickesh.patel@biol.lu.se).

Materials availability

This study did not generate any new or unique reagents.

Data and code availability

The data from this manuscript are published as an experiment entry in the InsectBrainDatabase.⁶⁰ Identification numbers are listed in the [key resources table](#).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Commercial hives of *Bombus terrestris* (Natupol Smart, Koppert Biological Systems) were shipped to Lund University and were housed at room temperature under a 12:12 light:dark cycle. One day after arrival, the wings of all individuals were clipped. One day following clipping, the hive was introduced into the experimental arena and bees were allowed to freely forage. They were allowed access to a feeder with 1M sugar-water solution and ground pollen (Bipollen EKO, Rawpowder Sweden AB) placed at the center of the arena. Foraging bees were labelled with unique small colored tags adhered to the back of the thorax. Data were collected from 91 individual bees from three hives, with each individual only being used once per experiment.

METHOD DETAILS

Experimental Arenas

Two relatively featureless, 1.5 m-diameter circular navigation arenas were constructed with a white composite wood base and were elevated 1.2 m to allow access below the arena (Figures 1A, 1B, and S1). The first arena contained a hive entrance 25 cm from the arena's periphery. The hive was placed beneath the arena and was connected to the arena by a tube with three doors to have control of bees that wished to enter the arena (Figure S1D). Sugar-water and pollen were placed in a removable conical feeder at the center of the arena. The conical feeder had a door which could slide over the bee while it was feeding, trapping it in the feeder (Figure S1C). Both the feeder and the hive-entrance were placed below ground-level of the arena so they were not visually detectable by bumblebees while in the arena. Trials were recorded from above using a raspberry pi camera connected to a raspberry pi unit (Raspberry Pi Foundation) outside of the arena, where experiments were observed and trials recorded.

The second arena was identical to the first with the exception that no hive or hive entrance was present and two potential locations were present where a conical feeder could be placed (at the center and near the edge of the arena). When a feeder was not occupying one of these two potential feeder locations, a tight plug, level with the arena floor and constructed of the same material as the base of the arena was present instead.

Arenas were placed in a dark room and surrounded by thick matte black curtains and lit from above using a centered, diffused, custom built LED light source with sets of ultraviolet (40 units, LZ1-00UV0R, LED Engin) and white LEDs (37 units, LZ1-10CW02, LED Engin). Composite filters constructed of a linear polarizer (38% transmission neutral grey, Rosco Laboratories) and two sheets of tracing paper were placed under each light source. When the polarizer side of the composite sheet faced downwards towards the arena, light was linearly polarized to an average degree of 99.78% from 350 to 600 nm (98.23% from 350 to 450 nm). For unpolarized fields, the depolarizing waxed paper side faced downwards, reducing the average degree of polarization to 2.32% from 350 to 650 nm (1.85% from 350 to 450 nm) (Figure 1D). The overhead polarization stimulus had an angular diameter of 22.6° of when viewed at ground level from the center of the arena and could be rotated up to 90° from outside of the arena. Four green LED (LZ4-00G108, LED Engin) point sources were positioned at 90° intervals around the arena with one at a 22.5° offset from the direction of the hive entrance from the center of the arena at an elevation of 21.8° above ground level when viewed from the center of the arena. All four LEDs could be controlled from outside the arena. Since the long-wavelength photoreceptor spectral-class in bumblebees provides achromatic contrast,³⁴ green LEDs were chosen to create a bright point source. Irradiance spectra of all light sources can be reviewed in Figure 1C.

Spectrometry

Irradiance measurements were taken from the center of the arenas at ground level using an Ocean Optics USB2000 spectrometer (Ocean Optics) connected to a 400 μ m diameter, fiber-optic cable with a cosine-correcting head. Irradiance measurements of the green LED point source were taken with the cosine-corrector angled towards the light source at a distance 20 cm from it (Figure 1C).

Transmittance measurements of the polarizing filter were taken using the same spectrometer system without a cosine-correcting head. The percent polarization of light transmitted through the filter was determined by calculating the ratio of the percent transmission of the polarizing filter layered with a second polarizer oriented first perpendicularly and then parallel to the first filter, both with the polarizing and then the depolarizing side of the filter facing upwards (Figure 1D).

Hive Observations

A hive containing bees with clipped wings were introduced into the experimental arena and were allowed to freely forage with a polarized light source present overhead and a feeder with 1M sugar-water and pollen at the center of the arena. During the first day, each individual bee that emerged from the hive was labeled with colored tag. During the following two days, the number of times each bee made a foraging trip from the hive to the feeder and back within the hour was recorded over the course of the full 12-hour light-cycle. The foraging activity of the hive was recorded one day per week for an additional three weeks following the initial two observation days. Any bees emerging from the hive without an identity tag following the initial week had their wings clipped, were given a tag, and were assigned to a new cohort.

Experimental Procedures

A hive containing bees with clipped wings were introduced into the experimental arena and were allowed to freely forage for three days. During familiarization, a feeder with sugar-water and pollen was placed at the center of the arena. During this period, either the overhead polarized light field (before polarization orientation, displacement experiments, and behavioral resilience experiments), the green LED point source (before green point-source orientation experiments), or both light sources (before conflict experiments) were present.

After familiarization, the doors to the hive were closed and experiments were conducted. During experiments, bees were let out of the hive individually, allowed to locate the feeder, and were trapped in the feeder. During this time, any experimental manipulations were performed before the bees were released and allowed to return to the hive. All conditions per experiment were performed in a randomized order.

During experiments in which animals were displaced, once bees were trapped in the feeder, the entire feeder was removed from the underside of arena 1 and placed at either the center or edge position in arena 2. Bees were then released and allowed to leave the

feeder. Once they returned to the feeder, they were trapped in the feeder once more, displaced back to arena 1, and allowed to return back to the hive (Figure 3A). To control for any effect the edge of the plug filling the center feeder position during edge-displacement experiments may have had on influencing a homing bee to end its home-vector and start a search pattern, edge displacement experiments were repeated when the polarized light field over the second arena was offset 45° from the polarized light field in the first arena, orienting homing bees away from the center of arena and the plug that was located there.

During experiments in which the polarization field was manipulated, once an experimental animal found the central feeder, the bee was trapped and the composite polarization filter with the polarizer-side facing downwards over the arena, creating a linearly polarized field, was rotated 90° from its original position and the bee was released. In control trials, the polarizer was rotated 45° forward and backwards by the researcher, so it ended at its original position (Figure 4A). To verify that bees were orienting to the polarization pattern the filter produced, the filter was placed diffuser-side downward over the arena, creating a depolarized field during the entire trial and was rotated 45° forward and backwards so it ended at its original position.

During green point-source orientation experiments, trials were run in the presence of a single green LED 157.5° relative to the hive entrance from the center of the arena. The position of the LED was chosen to limit parallax when a bee was moving towards or away from the feeder without having the LED completely aligned with the hive axis, making the cue appear as an infinitely distant one would while preventing the use of a simple phototaxis strategy to follow the exact home direction. Once a bee was trapped at the central feeder, the green LED was turned off and an identical one was turned on 180° from the original LED's position and the bee was released. As a control, the green LED was turned off and turned back on at its original position when a bee was trapped at the feeder.

During conflict experiments, both the overhead polarization field and the green LED was present. Once an animal was trapped at the feeder, one of four conditions were enacted: 1. The polarizer was rotated 45° forward and backwards and the green LED was turned off and back on at the original position (neither cue's position was changed). 2. Both cues were rotated 90° in the same direction. 3. The polarizer was rotated 90° and the green LED remained in the same position. 4. The green LED was rotated 90° and the polarizer remained in the same position.

During anesthetization and surgery experiments, once a bee found the central feeder, it was trapped and removed from the arena. The bee was then placed in a -20°C freezer for between 7 and 15 minutes, until it could no longer move any of its limbs (a chill-coma). The bee was then removed from the freezer and three shallow slits in a U-shape were cut into its head cuticle over the location of the cerebral ganglion (Figure 7A). The cuts were patched with a light-curing dental resin (tgflow, Technical & General Ltd.) and the bee was allowed to recover in an upright tube (35 mm inner diameter) in the dark for six hours. Following recovery, bees were placed back in the conical feeder and returned to the center of the arena, with the overhead polarized light field either oriented the same or 90° from its original position pre-surgery and homeward paths were observed. The same procedure was used for chill-coma anesthesia experiments, where bees were placed into a chill-coma and immediately returned to the arena without undergoing a surgery. For the 12-hour time-delay experiments, bees were trapped at the feeder for 12 hours without further manipulations and returned to the arena.

QUANTIFICATION AND STATISTICAL ANALYSIS

Foraging paths to and from food locations to the hive were video recorded at 10 frames per second. Videos were unwarped to correct for lens distortion and distortion due to camera placement using the Camera Calibration Toolbox in Matlab (vR2020a, MathWorks). To differentiate homeward paths from continued arena exploration, paths from the feeder were considered to be homeward paths when a straight path started at most 14 cm from the feeder and when they did not deviate more than 90° from their initial trajectories for at least 28 cm. From these homeward paths, search behaviors were determined to be initiated when an animal turned more than 90° from its initial trajectory and continued walking at a distance of at least 14 cm. The number of trials with homeward paths per experimental condition is reported in Table S3.

Paths were automatically tracked using the software ivTrace (v2.23, Bielefeld University), from which the output is given as Cartesian coordinates. From these measures, home vector lengths and beeline distances from the feeder to the hive entrance were determined. The straightness of paths and mean walking speeds during both the home vector and search phases of homeward travel were also calculated. Further, for search behaviors from edge displacement experiments without an offset in polarizer orientation between the two arenas, mean walking speeds binned over one second increments (10 frames) were calculated for the entirety of the search.

Additionally, the orientation of homeward paths when bees were 28 cm from the feeder in relation to the hive entrance was recorded using the angle tool in ImageJ (v2.3.0, Fiji).

Search behaviors with at least one completed loop were analyzed from all trials when animals were displaced to the edge position of arena 2 with the polarizer set in the same position between the two arenas, since searches should start at approximately the center of the arena, negating the potential effect the proximity to the arena wall would have on biasing the geometry of the search. The radii of these search behaviors were measured as the farthest distance of each successive increasing search loop from the point of search initiation (i.e. the end point of the home vector) using the line tool in ImageJ. To limit the potential effect the proximity to the arena wall may have had on search expansion, the absolute radius of each search loop was measured from the beginning of the search until bees reached a distance of 25 cm from the arena wall. To observe the general expansion pattern of the maximal search radius, only search loops larger than previous loops were included. These values were plotted over time and fitted with a power function.

To remove the potentially distorting impact of a single, very long search, this analysis was repeated for only the initial 100 seconds of each search. This further limited the influence of the arena wall on free search expansion.

All statistical analyses were run on R (v4.0.2, R Core Development Team) with the “autoimage”, “circular”, “ggplot2”, “plotrix”, and “shape” plugins. All statistical analyses for experiments conducted under a polarized light field were performed after using the doubling angles procedure for bimodal data outlined in Batschelet.⁶¹ Orientation data were analyzed using the following procedures for circular statistics⁶¹: Rayleigh tests of uniformity were used to determine if homeward paths were oriented within a group for all experiments. Watson Two-Sample tests for homogeneity were used to determine if group orientations were significantly different from one another. All reported mean values for orientation data are circular means and circular standard deviations.

For experiments with the point source present, V-tests for orientation in respect to an expected direction were performed to determine if the observed homeward path orientations better matched the predicted direction of the hive or the direction opposite of the LED present in the arena during those experiments by comparing the test-statistic outputs of the tests.

Kruskal-Wallis tests were used to evaluate differences in the path straightness and walking velocity during the home-vector and search phases of homeward paths during displacement experiments. If significant differences were observed, post-hoc pairwise Wilcoxon ranked tests were used to determine which specific groups were different from one another.

Bonferroni corrections were used for all tests when applicable. All statistical information including sample sizes, test statistics, p values, means, and standard deviations are presented in [Tables S1–S4](#).